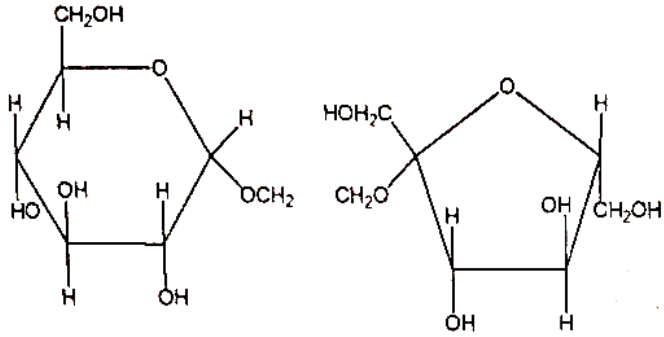
Biomolecules

1. Which of the following pairs give positive Tollen's test?

1) Glucose, sucrose
2) Glucose, fructose
3) Hexanal, acetophenone
4) Fructose, sucrose

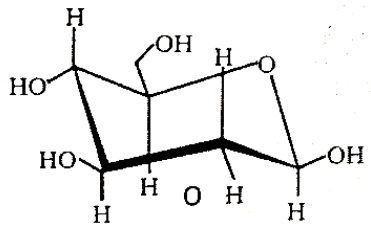
2. The correct statement about the following disaccharide is



(a) **(b)**

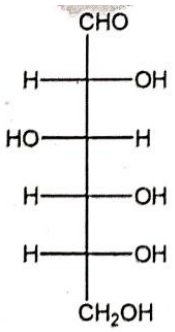
1) Ring (a) is pyranose with $\alpha-$glycosidic link
2) Ring (a) is furanose with $\alpha-$glycosidic link
3) Ring (b) is furanose with $\alpha-$glycosidic link
4) Ring (b) is pyranose with $\beta-$glycosidic link

3. The following carbohydrate is

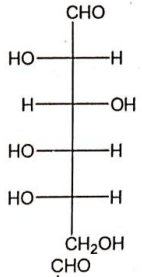


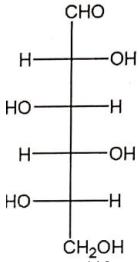
1) A ketohexose
2) An aldohexose
3) An α-furanose
4) An α-pyranose

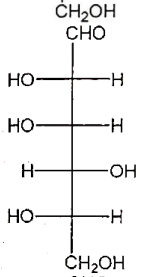
4. The structure of D-(+)-glucose is

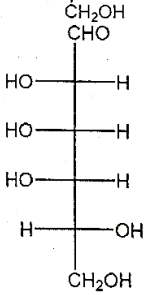


The structure of L-(-)-glucose is

1) 

2) 

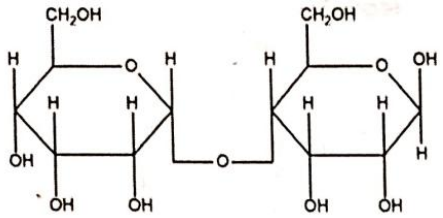
3) 

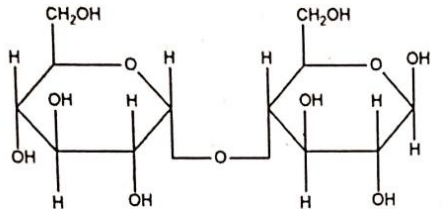
4) 

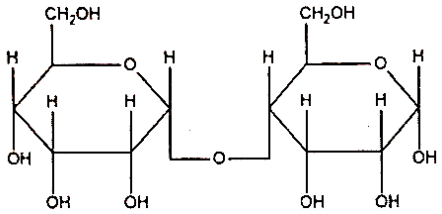
5. Glucose on prolonged heating with HI gives

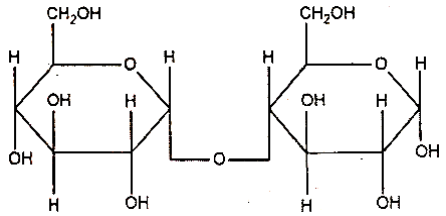
1) 1-Hexene
2) Hexanoic acid
3) 6-iodohexanal
4) n-Hexane

6. Which of the following is correct structure of α-anomer of maltose?

1) 

2) 

3) 

4) 

7. Hydrolysis of sucrose gives

1) α -D-(+)-Glucose and
   α -D-(-)-Fructose
2) α -D-(+)-Glucose and
   β -D-(-)-Fructose
3) α -D-(-)-Glucose and
   β -D-(-)-Fructose
4) α -D-(-)-Glucose and
   α -D-(+)-Fructose

8. Given below are two statements:

Statement I: Maltose has two $\alpha-D-$ glucose units linked at C_1 and C_2 and is a reducing sugar.

Statement II: Maltose has two monosaccharides: $\alpha-D-$glucose and $\beta-D-$ glucose linked at C_1 and C_6 and it is a non-reducing sugar. In the light of the above statements, choose the correct answer from the options given below.

1) Both Statement I and Statement II are true
2) Both Statement I and Statement II are false
3) Statement I is true but Statement II is false
4) Statement I is false but Statement II is true

9. Animal starch is the other name of

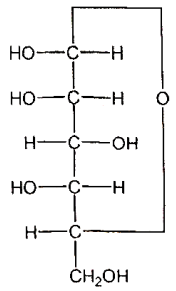
1) Amylose
2) Maltose
3) Glycogen
4) Amylopectin

10. Match List I with List II.

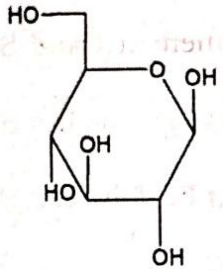
List I		List II	
(A)	Glucose + HI	(I)	Gluconic acid
(B)	Glucose + Br_2 water	(II)	Glucose pentaacetate
(C)	Glucose + Acetic anhydride	(III)	Saccharic acid
(D)	Glucose + HNO_3	(IV)	Hexane

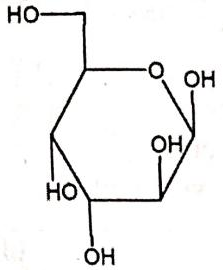
1) A-IV, B-I, C-II, D-III
2) A-IV, B-III, C-II, D-I
3) A-III, B-I, C-IV, D-II
4) A-I, B-III, C-IV, D-II

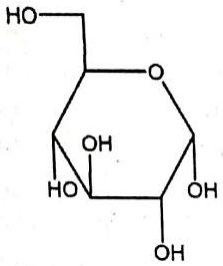
11. For the below given cyclic hemiacetal (X), the correct pyranose structure is:

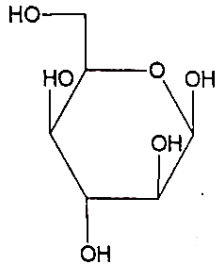


(X)

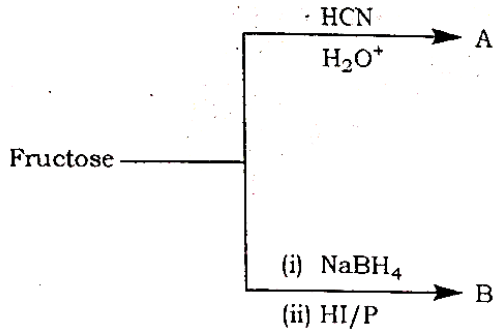
1) 

2) 

3) 

4) 

12. The formulas of A and B for the following reaction sequence are:

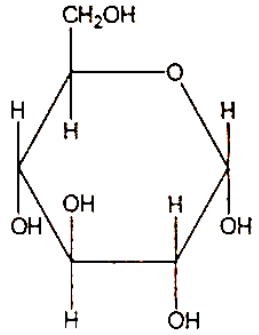


1) $A = C_7H_{14}O_8, B = C_6H_{14}$
2) $A = C_7H_{13}O_7, B = C_7H_{14}O$
3) $A = C_7H_{12}O_8, B = C_6H_{14}$
4) $A = C_7H_{14}O_8, B = C_6H_{14}O_6$

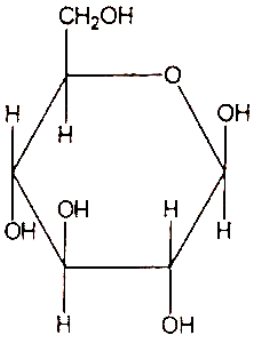
13. Match items of Row I with those of Row II

Row I:

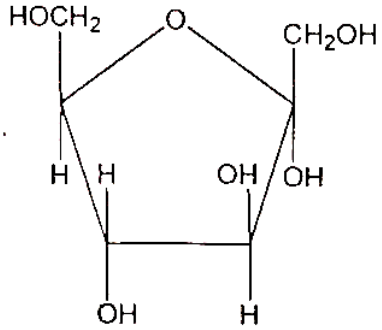
(P)



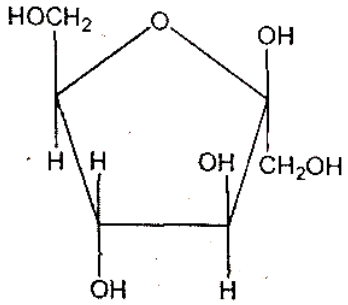
(Q)



(R)



(S)



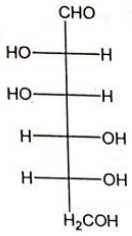
Row II:

(I) α –D–(–) Fructofuranose
(II) β –D–(–) Fructofuranose
(III) α –D–(–) Glucopyranose
(IV) β –D–(–) Glucopyranose

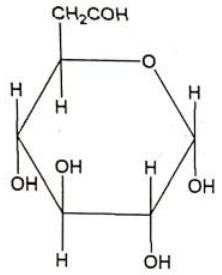
Correct match is:

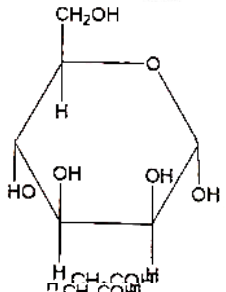
1) P-IV, Q-III, R-I, S-II
2) P-I, Q-II, R-III, S-IV
3) P-III, Q-IV, R-II, S-I
4) P-III, Q-IV, R-I, S-II

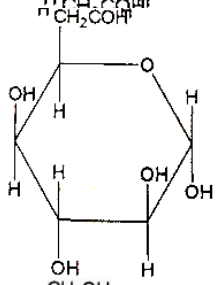
14. The correct representation in six membered pyranose form for the following sugar [X] is

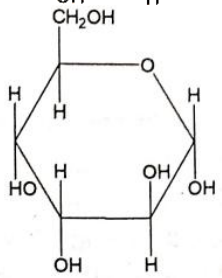


Sugar [X]

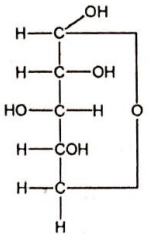
1) 

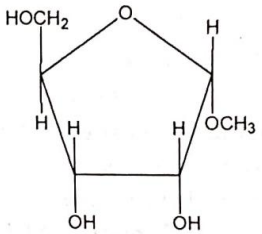
2) 

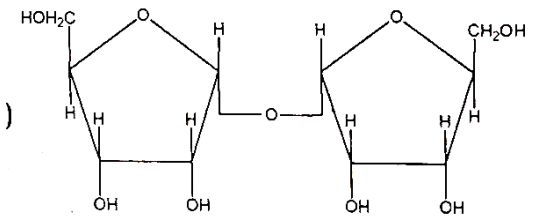
3) 

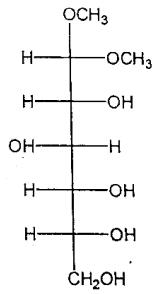
4) 

15. Which of the following is reducing sugar?

1) 

2) 

3) 

4) 

16. Glucose does not react with

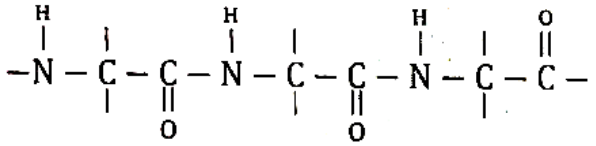
1) Br_2 / H_2O
2) H_2NOH
3) HI
4) $NaHSO_3$

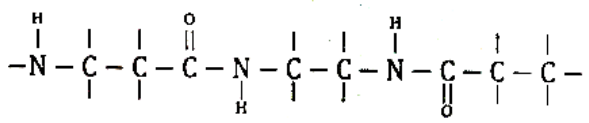
17. Assertion: Proteins on hydrolysis produce amino acids.

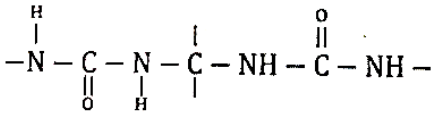
Reason: Amino acids contain $-NH_2$ and -COOH groups.

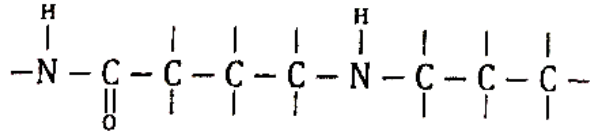
1) If both Assertion and Reason are true and the Reason is the correct explanation of the Assertion.
2) If both Assertion and Reason are true and but Reason is not the correct explanation of the Assertion.
3) If Assertion is true but Reason is false.
4) If the Assertion and Reason both are false.

18. Which one of the following structures represents the peptide chain?

1) 

2) 

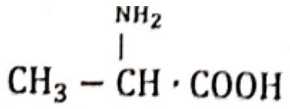
3) 

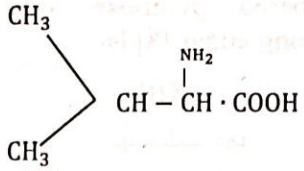
4) 

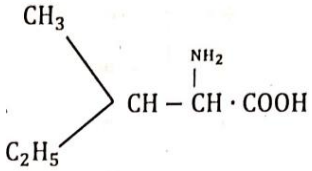
19. Haemoglobin is

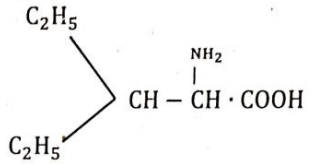
1) An enzyme
2) A globular protein
3) A vitamin
4) A carbohydrate

20. The structural formula of an amino acid, isoleucine is

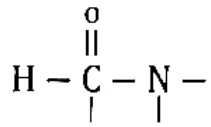
1) 

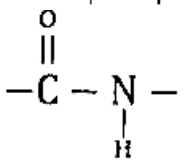
2) 

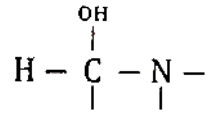
3) 

4) 

21. Which one is the correct representation of peptide bond

1) 

2) 

3) 

4) None of these

22. Amino acids usually exist in the form of Zwitter ions. This means that it consists of

1) The basic group -NH_2 and the acidic group -COOH
2) The basic group $-NH_3^+$ and the acidic group $-CO_2$
3) The basic group $-CO_2^-$ and the acidic group $-NH_3^+$
4) No acidic or basic group

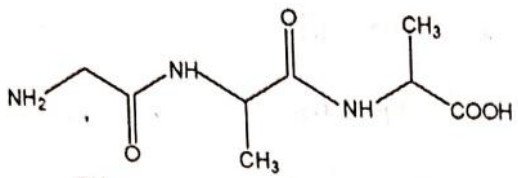
23. A nanopeptide contains _____ peptide linkages

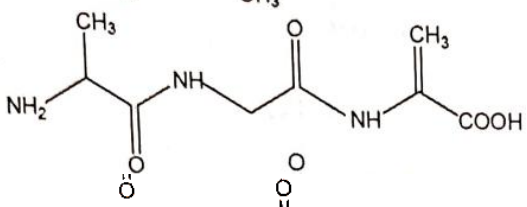
1) 10
2) 8
3) 9
4) 18

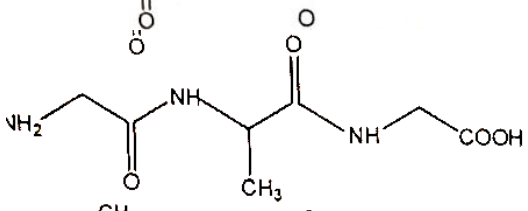
24. At pH = 4, glycine exists as

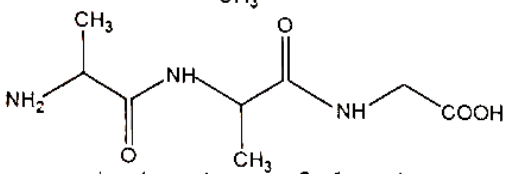
1) $H_3\overset{+}{N}-CH_2-COO^-$
2) $H_3\overset{+}{N}-CH_2-COOH$
3) H_2N-CH_2-COOH
4) $H_2N-CH_2-COO^-$

25. A tripeptide is written as GlycineAlanine-Glycine. The correct structure of the tripeptide is

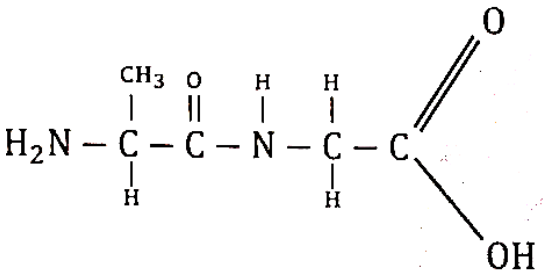
1) 

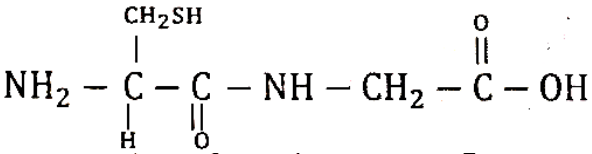
2) 

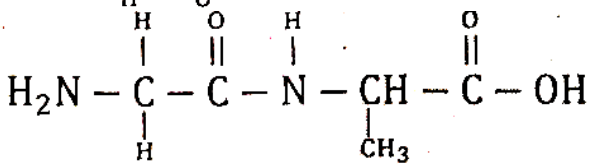
3) 

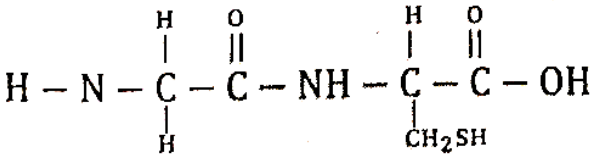
4) 

26. The correct structure of the dipeptide gly-ala is

1) 

2) 

3) 

4) 

27. Simplest biological amino acid is

1) Alanine
2) Lysine
3) Histidine
4) Glycine

28. Assertion: All Amino acids exist as Zwitter ions.

Reason: Amino acids have both -NH_2 and -COOH group.

1) If both Assertion and Reason are true and the Reason is the correct explanation of the Assertion.
2) If both Assertion and Reason are true and but Reason is not the correct explanation of the Assertion.
3) If Assertion is true but Reason is false.

4) If the Assertion and Reason both are false.

29. The difference between amylose and amylopectin is

1) Amylopectin have $1 \rightarrow 4$ $\alpha-$linkage and $1 \rightarrow 6$ $\alpha-$ linkage
2) Amylose have $1 \rightarrow 4$ $\alpha-$linkage and $1 \rightarrow 6$ $\beta-$linkage
3) Amylopectin have $1 \rightarrow 4$ $\alpha-$linkage and $1 \rightarrow 6$ $\beta-$linkage
4) Amylose is made up of glucose and galactose

30. Which of the following compounds can form a zwitterion?

1) Aniline
2) Acetanilide
3) Benzoic acid
4) Glycine

Answer Key

1	2	**16**	4
2	1	**17**	2
3	2	**18**	1
4	1	**19**	2
5	4	**20**	3
6	4	**21**	2
7	2	**22**	3
8	3	**23**	2
9	3	**24**	2
10	1	**25**	3
11	4	**26**	3
12	1	**27**	4
13	4	**28**	1
14	2	**29**	1
15	1	**30**	4